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Revision 2021 • Semester 2

Production Engineering - I

CODE: 2111 **CREDITS: 3** **TOOL & DIE ENGINEERING**

Teaching Scheme

L: 3 | T: 0 | P: 0

Total: 45 Hours / Semester

Assessment

CIE: 2 Series Tests + Attendance

ESE: End Semester Exam

Prerequisites

Engineering Graphics (Sem 1)

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Course Outcomes (COs)

CO No.	Description	Hours	Cognitive Level
CO 1	Demonstrate the principles of mechanics of metal cutting and tool life	13	Applying
CO 2	Illustrate types of cutting tool materials, importance of tool geometry and use of cutting fluids	10	Applying
CO 3	Demonstrate bench work and fitting, welding and forging	10	Applying
CO 4	Describe the types of patterns, moulds, and powder metallurgy	10	Understanding

Module 1

Mechanics of Metal Cutting & Tool Life

▼ 1.1 Mechanics of Metal Cutting

Metal cutting is the process of removing unwanted material from a workpiece in the form of chips using a wedge-shaped tool. The primary mechanism is **Plastic Deformation** in the shear zone.

Types of Metal Cutting Processes:

- **Orthogonal Cutting:** The cutting edge is perpendicular to the direction of tool travel. It is a 2D cutting process.
- **Oblique Cutting:** The cutting edge is inclined at an angle to the direction of tool travel. It is a 3D cutting process.

Types of Chips:

Continuous Chips: Formed during cutting of ductile materials at high speeds. Good surface finish.

Discontinuous Chips: Formed during cutting of brittle materials (like Cast Iron) or ductile materials at very low speeds.

Built-up Edge (BUE): Formed when high friction at the tool-chip interface causes chip material to weld to the tool tip.

▼ 1.2 Tool Wear & Tool Life

Tool wear is the gradual loss of tool material during the cutting process. The two main types are:

- **Flank Wear:** Occurs on the relief face of the tool due to rubbing against the machined surface.
- **Crater Wear:** Occurs on the rake face of the tool due to the flow of chips.

Tool Life: The duration of actual cutting time between two successive tool grinds.

Formula: `V * T^n = C`

Cutting Speed (V) in m/min: 100

Exponent (n): 0.125

Constant (C): 200

Calculate Tool Life (T)

Result: -- minutes

Module 2

Tool Materials, Geometry & Cutting Fluids

▼ 2.1 Cutting Tool Materials

A tool material must have High Hardness, Hot Hardness, Toughness, and Wear Resistance.

Material	Characteristics	Applications
High Speed Steel (HSS)	Good toughness, can be resharpened	Drills, Reamers, Milling cutters
Cemented Carbides	High hot hardness, high speed	Lathe inserts, high-production tools
Ceramics	Extreme heat resistance, brittle	Finish machining of hard alloys
Diamond	Hardest known material	Non-ferrous metals, precision finishing

▼ 2.2 Single Point Tool Geometry

The performance of a tool depends on its angles:

- **Back Rake Angle:** Controls chip flow and tool strength.
- **Side Rake Angle:** Directs chips away from the finished surface.
- **End Relief Angle:** Prevents tool flank from rubbing against the workpiece.
- **Side Relief Angle:** Prevents side of tool from rubbing.

Module 3

Bench Work, Welding & Forging

▼ 3.1 Fitting & Bench Work

Bench work involves operations performed by hand at a workbench using tools like files, hacksaws, and chisels.

- **Vices:** Bench vice, Machine vice, Pipe vice.
- **Files:** Classified by length, shape (Flat, Round, Square), and grade of cut (Bastard, Smooth).
- **Marking Tools:** Scribers, Punches, V-block, Angle plate.

▼ 3.2 Welding Processes

Welding is a fabrication process that joins materials, usually metals, by using high heat to melt the parts together.

Arc Welding

Uses an electric arc to create heat. Types: SMAW (Shielded Metal Arc), TIG (Tungsten Inert Gas), MIG (Metal Inert Gas).

Gas Welding

Uses combustion of oxygen and acetylene. Flames: Neutral, Oxidizing, Carburizing.

Module 4

Patterns, Moulding & Powder Metallurgy

▼ 4.1 Patterns & Moulding

A **Pattern** is a replica of the object to be cast, used to prepare the mould cavity.

- **Pattern Types:** Solid piece, Split piece, Loose piece, Match plate, Sweep pattern.
- **Moulding Sand Properties:** Porosity, Flowability, Collapsibility, Adhesiveness.

▼ 4.2 Powder Metallurgy

Powder Metallurgy (PM) is the process of blending fine powdered materials, pressing them into a desired shape (compacting), and then heating the compressed material in a controlled atmosphere to bond the material (sintering).

1. Powder Production (Atomization, Reduction)
2. Blending/Mixing
3. Compacting (Pressing)
4. Sintering (Heating below melting point)
5. Secondary operations (Sizing, Infiltration)

Formula Bank & Calculations

1. Cutting Speed (V)

```
V = (π * D * N) / 1000
```

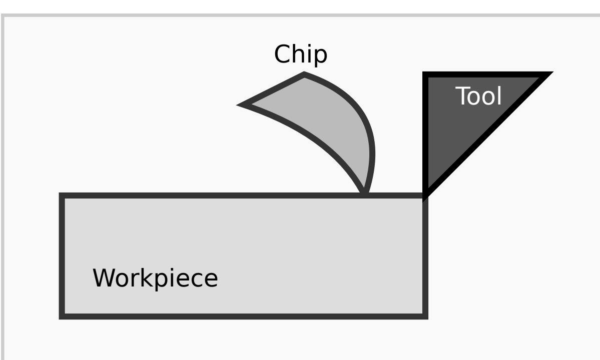
Where:
D = Diameter of workpiece (mm)
N = Spindle speed (RPM)
V = Cutting speed (m/min)

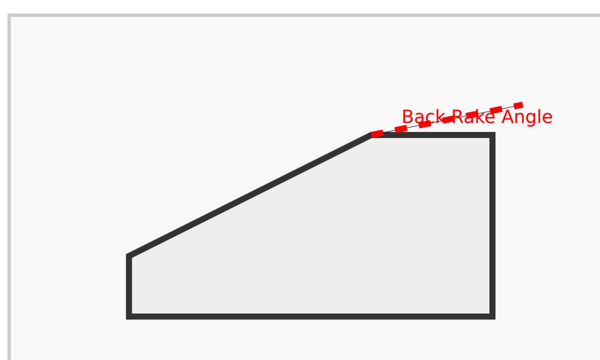
2. Taylor's Tool Life

```
V * T^n = C
```

Where:
T = Tool life (min)
n = Tool life exponent
C = Machining constant

Visual Library





Basic Mechanics of Orthogonal Metal Cutting

Single Point Cutting Tool Geometry (Simplified)

Module-wise Practice Questions

Module 1 & 2

▶ **Q1: Differentiate between Orthogonal and Oblique cutting. (5 Marks)**

Module 3 & 4

▶ **Q2: Explain the steps in Powder Metallurgy. (5 Marks)**

Practice Model Paper

Part A (Short Answer - 2 Marks each)

- Define Machinability.
- What is a Built-up Edge?
- List two properties of moulding sand.

Part B (Medium Answer - 5 Marks each)

- Explain the functions of cutting fluids.
- Describe the different types of welding flames.
- Explain the advantages of hot working over cold working.

Part C (Long Answer - 10 Marks each)

- Explain Single Point Tool Geometry with a neat sketch.
- Describe various pattern allowances used in casting.

Quick Revision

Tool Life

High speed = Low tool life. Taylor's equation is key for exam problems.

Welding

TIG uses non-consumable tungsten; MIG uses consumable wire electrode.

Casting

Draft allowance is provided to remove the pattern easily from the sand.

Glossary

- **Machinability:** The ease with which a material can be cut.
- **Sintering:** Heating powder compacts to join particles without melting.
- **Rake Angle:** The angle between the tool face and a line perpendicular to the machined surface.
- **Flux:** Material used in welding to prevent oxidation.

Official Source Declaration

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